Stat 135 Section

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How is everyone doing?

Figure: *I'm so excited.* Source: `memegenerator.net`

A joke

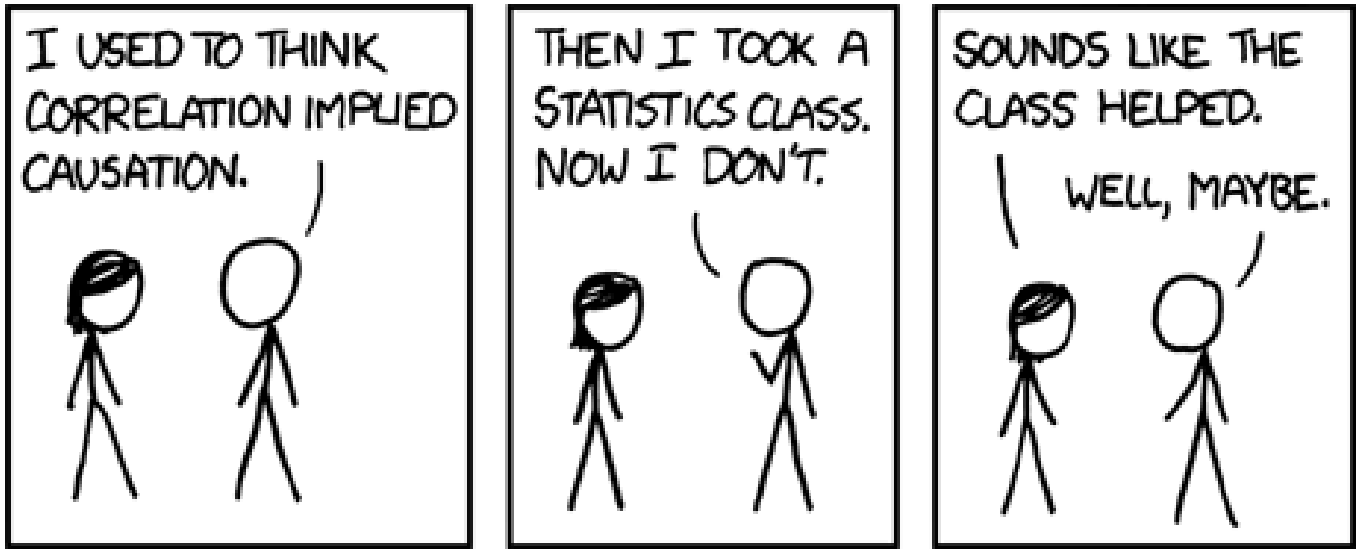

Figure: *Correlation.* Source: xkcd (`https://xkcd.com/552/`)

1A. Combining standard deviations

A population consists of n men and n women. The heights of the men have an average of μ_m and an SD of σ_m. The heights of the women have an average of μ_w and an SD of σ_w. Find a formula for the SD of the heights of all $2n$ people, in terms of μ_m, μ_w, σ_m and σ_w.

1B. Comparing samples

Two populations are surveyed with simple random samples. A sample of size n_1 is used for population I, which has a population standard deviation σ_1; a sample of size $n_2 = 2n_1$ is used for population II, which has a population standard deviation $\sigma_2 = 2\sigma_1$. Ignoring finite population corrections, in which of the two samples would you expect the estimate of the population mean to be more accurate?

2. Method of Moments (MoM)

Consider the i.i.d. random variables $X_1, X_2, \ldots, X_n \sim N(\mu, \sigma^2)$ with both μ and σ^2 unknown. Find $\hat{\mu}_{MM}$ and $\hat{\sigma}^2_{MM}$, i.e., the MoM estimators of the mean and the variance.

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3. Formula for variance

Let $x_1, x_2, \ldots, x_n$ be a list of numbers with mean μ and SD σ. Show that

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \mu^2. \tag{1}$$

R programming exercise

1. Go to Bcourses page, Pages, lab, and download `lab2.Rmd`.
2. Open file in RStudio.

Resources to learn bootstrap

- (Watered-down explanation of theory) Chapter 8.1-8.3 of Larry Wasserman, *All of statistics: a concise course in statistical inference*, Springer (2004).
- (Example-driven exposition) Lecture notes from Zhou Fan's Stat 200, taught at Stanford (`http://web.stanford.edu/class/stats200/Lecture19.pdf`)